
Hierarchical Lateral Inhibition Enables Robust Winner-Take-All Computation in Large-Scale Cortical Models

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Abstract

In computational neuroscience, models of visual cortex have long captured classical receptive field properties, yet the circuit mechanisms underlying extra-classical surround suppression remain unresolved. In vivo experiments have consistently shown a winner-take-all (WTA) dynamic, where a strongly stimulated cortical column suppresses neighboring activity by over 75%; this dynamic is essential for basic visual functions like figure-ground segmentation. Classical models attribute this computation to lateral inhibition within V1, yet prior work has struggled to reproduce the magnitude of suppression observed physiologically using models of V1 circuits alone. In this work, we hypothesized that robust WTA emergence requires extending lateral inhibitory competition to higher cortical areas, beyond V1. To test this hypothesis, we simulated a large-scale spiking neural network based on the connectivity of the canonical cortical microcircuit and systematically compared three different architectures: (1) V1 columns with lateral inhibition restricted to V1, (2) a hierarchical V1-V2 model with simple feedback between V1 and V2 but no lateral inhibition in V2, and (3) a hierarchical model with feedback where both V1 and V2 incorporate lateral inhibitory connections. Our experiments show that only when lateral competition is extended to V2 do the circuits generate a physiologically accurate surround suppression effect, achieving a suppression index of 75% that quantitatively matches in vivo recordings. In this architecture, V2 lateral inhibition gates the excitatory feedback returning to V1, enabling the network to dynamically encode relative stimulus strength even when absolute input remains constant. Our work explains a long-standing physiological puzzle and establishes that lateral inhibition in higher cortical areas is necessary to drive the circuit from a regime of general divisive normalization into a robust winner-take-all (WTA) state, which is essential for strong surround suppression. Moreover, this hierarchical inhibitory architecture provides a blueprint for designing systems that require context-dependent information processing.

1 Introduction

From a computational point of view, the contemporary understanding of sensory processing in the visual system is that the cortex is organized as a hierarchy of filters where neurons extract progressively complex features of the visual scene through stage-by-stage processing. Central to this conception is the classical receptive field (CRF), operationally defined as the specific region of visual space where the presence of a stimulus, such as light or a dark bar, can evoke a direct neuronal response, increasing the neuron's firing rate. In this classical model, processing has been interpreted as a local phenomenon, where the response of each functional unit is determined almost exclusively by the physical properties of the stimulus and its location within the field of view.

37 However, the strictly local view of neuronal processing has been challenged by the discovery of
 38 extra-classical receptive field (eCRF) effects. In this phenomenon, sensory responses are not only
 39 determined by stimuli within the CRF but are also robustly modulated by contextual information
 40 from the surrounding visual field [1]. The most prominent of these effects is surround suppression,
 41 a non-linear phenomenon in which expanding a stimulus beyond the CRF leads to a substantial
 42 decrease in firing rates. This implies a high level of global integration, challenging the conception
 43 of primary visual cortex as merely a set of isolated local filters, and suggests that the functional
 44 architecture of the visual system must incorporate mechanisms capable of integrating information
 45 across disparate spatial locations [2]. Long-range horizontal connections within V1 were initially
 46 proposed as the primary substrate for eCRF effects; however, accumulating evidence indicates that
 47 these lateral pathways are physiologically limited in both conduction velocity and spatial reach [3, 4].
 48 This has led to the prevailing hypothesis that feedback projections from higher-order areas, such as
 49 V2, play a decisive role in integrating global information, providing the necessary spatial extent and
 50 conduction speed to modulate V1 activity effectively [5].

51 Despite significant advances in developing biologically plausible cortical microcircuits, such as the
 52 canonical model proposed by Potjans and Diesmann [6], a substantial gap remains between theoretical
 53 predictions and experimental observations of eCRF effects. Most existing computational models fail
 54 to quantitatively replicate the magnitude of surround suppression observed physiologically, which
 55 can reach a reduction of approximately 75% in neuronal firing rates [7, 8]. This discrepancy often
 56 stems from feedback implementations that are either linear or excessively simplified. By neglecting
 57 the laminar architecture of the cortex and the specificity of inter-areal connections, these models fail
 58 to capture the intrinsic competitive dynamics of the visual hierarchy.

59 In this study, we hypothesized that robust surround suppression is not merely a byproduct of hi-
 60 erarchical feedback but an emergent property of competitive dynamics within the visual system.
 61 Utilizing the NEST simulator [9], we implemented a large-scale spiking neural network model
 62 based on the canonical cortical microcircuit to examine how inter-areal interactions shape eCRF
 63 effects. Specifically, we propose that lateral competition within higher-order areas, particularly V2, is
 64 the essential mechanism for regulating top-down modulation and achieving the suppression levels
 65 observed experimentally. By comparing multiple architectural configurations, we aim to demonstrate
 66 that inter-areal competition is the missing component required to replicate the non-linear contextual
 67 dynamics of the primary visual cortex.

68 Our main contributions are as follows. First, we systematically compare three hierarchical architec-
 69 tures of large-scale spiking cortical microcircuits, isolating the specific role of lateral inhibition in V2
 70 for extra-classical surround suppression. Second, we demonstrate that lateral competition within V2
 71 is necessary to achieve the 75% suppression index consistently reported in primate in vivo record-
 72 ings; architectures relying solely on V1 lateral connections or non-competitive V2 feedback both
 73 fail to replicate this magnitude. Finally, we show that this hierarchical winner-take-all mechanism
 74 enables the network to encode relative stimulus strength even when absolute input remains constant,
 75 revealing a computational principle directly transferable to artificial perception systems that require
 76 context-dependent normalization.

77 2 Related Work

78 Early computational approaches to extra-classical receptive field (eCRF) effects primarily emphasized
 79 the role of intrinsic horizontal connections within the primary visual cortex (V1) [10, 11]. These
 80 lateral projections enable spatial integration by linking neurons with non-overlapping receptive fields
 81 across the cortical surface. However, horizontal connections face significant physiological constraints
 82 in both spatial extent and conduction velocity, which limit their capacity to account for the rapid,
 83 long-range dynamics of surround suppression observed experimentally [3, 4]. Consistent with these
 84 biological constraints, simulations of lateral inhibition restricted to V1 yield only modest suppression,
 85 typically a 30–35% reduction in firing rate [12, 13], which is far below the 75% reduction reported
 86 in vivo. These findings suggest that local V1 circuitry alone lacks the dynamic range necessary to
 87 fully account for global contextual integration.

88 An alternative hypothesis proposes that feedback projections from higher-order areas, particularly V2,
 89 serve as the primary driver of long-range contextual modulation [5]. However, existing computational
 90 implementations of this feedback have largely relied on simplified loops that treat top-down signals

91 as linear additive inputs or non-specific gain modulators [14, 15]. Such formulations fail to capture
92 the highly non-linear and stimulus-specific nature of eCRF effects. Critically, no existing model has
93 incorporated lateral inhibitory competition within the feedback-generating higher-order areas—a
94 mechanism that may be essential for achieving the precise spatial and feature-specific filtering
95 observed physiologically. This gap motivates the present study: we investigate whether embedding
96 competitive dynamics within V2 feedback pathways can yield the robust surround suppression that
97 purely local or linearly integrated models have failed to reproduce.

98 Substantial progress has been made in developing biologically constrained spiking neural network
99 models of cortical processing. The canonical microcircuit model of Potjans and Diesmann [6] has
100 become a widely adopted framework for reproducing the laminar connectivity and asynchronous-
101 irregular dynamics of local cortical patches. This model has been successfully extended to multi-area
102 hierarchies [16] and applied to cognitive tasks such as working memory and decision-making [17].
103 Concurrently, spiking networks trained via surrogate gradients have achieved competitive performance
104 on visual recognition benchmarks [18], demonstrating that spike-based computation can scale to
105 complex tasks. However, these models typically treat hierarchical interactions as feedforward feature
106 extraction or simple convergent feedback, without incorporating lateral inhibitory competition within
107 higher areas. Whether the addition of such competitive dynamics would enable these networks to
108 exhibit the robust, stimulus-specific surround suppression characteristic of biological vision remains
109 an open question, which we address in this work.

110 The phenomenon of surround suppression has been extensively studied through the lens of divisive
111 normalization, a canonical neural computation in which a neuron’s response is divided by the
112 aggregate activity of a broader neuronal pool [15]. Early formulations by Heeger [19] demonstrated
113 that divisive normalization can account for contrast-dependent saturation and cross-orientation
114 suppression in V1 using simple computational rules. Subsequent work generalized this principle
115 to a wide range of sensory and cognitive phenomena [20, 21]. In parallel, the machine learning
116 community has independently converged on normalization as an essential architectural component.
117 Batch normalization [22], layer normalization [23], and divisive normalization networks [24] have
118 each shown that rescaling activations relative to local or global statistics improves training stability and
119 generalization. Importantly, these implementations operate through feedforward or self-normalizing
120 mechanisms within a single processing stage, and do not incorporate the inter-areal competitive
121 dynamics that characterize cortical hierarchies.

122 In this work, we propose a hierarchical competition mechanism to explain eCRF effects. This
123 mechanism is closely aligned with predictive coding frameworks [25, 26], in which higher cortical
124 areas generate top-down predictions that are compared against incoming sensory evidence from
125 lower levels; mismatches produce prediction errors that propagate upward, while accurately predicted
126 features are suppressed through a process known as *explaining away*. This computation bears
127 functional similarities to biased competition models of attention [27, 28], where top-down signals
128 resolve competition among multiple stimuli by selectively enhancing or suppressing specific feature
129 representations. The winner-take-all competition that we implement in V2 can be interpreted as
130 a circuit-level instantiation of these principles: when a peripheral stimulus drives a competing V2
131 module, it actively inhibits the center-aligned module’s feedback, effectively signaling that the central
132 activity is redundant given the broader spatial context. By grounding these abstract theories in a
133 concrete spiking microcircuit with laminar specificity, our work bridges normative computational
134 principles and their biologically plausible implementation.

135 In summary, none of these research lines, including biophysical models of cortical microcircuits,
136 divisive normalization in theory and practice, and predictive coding, explain how extra-classical
137 surround suppression emerges from hierarchical interactions. Specifically, the role of lateral inhibitory
138 competition in higher visual areas as a gating mechanism for top-down feedback remains unexplored
139 in large-scale spiking models. Our work addresses this gap by demonstrating that V2 lateral inhibition
140 is necessary, and within the architectures tested sufficient, to reproduce the hallmark signatures of
141 robust sensory normalization.

142 3 Methods

143 We implemented a large-scale spiking neural network using the NEST simulator [9]. We constructed
144 three architectural configurations of increasing complexity, all grounded in the canonical cortical

microcircuit [6]: (1) V1 columns with only lateral intra-areal connections, (2) a V1-V2 hierarchy with simple convergent feedback, and (3) a V1-V2 hierarchy with reciprocal lateral inhibition between V2 modules. In the following sections we describe the neuron model, the laminar microcircuit architecture, the hierarchical connectivity, the stimulation protocols, and the data analysis pipeline.

3.1 Neuron Model and Simulation Environment. Neurons were modeled as leaky integrate-and-fire (LIF) units with exponential-decaying postsynaptic currents. The subthreshold membrane potential dynamics are governed by the membrane time constant ($\tau_m = 10$ ms), membrane capacity ($C_m = 250$ pF), and resting potential ($V_{\text{rest}} = -65$ mV). A spike is emitted when V_m reaches the threshold ($V_{\text{th}} = -50$ mV), after which the potential is clamped to the reset value ($V_{\text{reset}} = -65$ mV) for an absolute refractory period ($\tau_{\text{ref}} = 2$ ms). Excitatory synaptic weights were set to $w = 87.8 \pm 8.8$ pA with a decay time constant $\tau_{\text{syn}} = 0.5$ ms, and relative inhibitory synaptic strength was fixed at $g = -4$. Transmission delays were drawn from uniform distributions of 1.5 ± 0.75 ms for excitatory synapses and 0.75 ± 0.38 ms for inhibitory synapses. Background cortical activity was emulated via Poisson spike trains delivered to all neurons at a mean rate of 8 Hz. All parameters were tuned to sustain the asynchronous-irregular firing regime characteristic of awake cortical states. Simulations were performed using the NEST simulator [9]. The membrane potential evolves according to

$$\tau_m \frac{dV_m}{dt} = -(V_m - V_{\text{rest}}) + R_m I_{\text{syn}}(t), \quad (1)$$

where R_m is the membrane resistance and $I_{\text{syn}}(t)$ is the total synaptic current. Each synaptic input generates a conductance-based current with exponential decay,

$$I_{\text{syn}}(t) = w e^{-t/\tau_{\text{syn}}}, \quad (2)$$

where w is the synaptic weight and τ_{syn} is the synaptic time constant.

3.2 Laminar Microcircuit Architecture. Each V1 module represents a 1 mm^2 patch of cortical tissue, based on the full-scale canonical microcircuit model proposed by Potjans and Diesmann [6]. It comprises approximately 77,000 neurons distributed across four laminar layers: L2/3, L4, L5, and L6. Within each layer, the network maintains a physiological ratio of excitatory to inhibitory neurons of approximately 80/20. Local recurrent connectivity follows empirically derived connection probabilities.

To capture intra-areal spatial integration, long-range horizontal projections within V1 are incorporated. These lateral connections are restricted to the superficial L2/3 layer and are functionally segregated by distance. Nearby connections link functionally similar adjacent columns with both excitatory and inhibitory synapses, with transmission delays of $d_{\text{n,lat}} = 5 \pm 1.5$ ms. Distant connections, in contrast, link columns with disparate orientation preferences and operate via a predominantly suppressive mechanism through disinaptic inhibition, with a slower conduction delay of $d_{\text{d,lat}} = 8 \pm 2.5$ ms.

(CD: Aca se habla del apéndice)The complete connection probability matrix between the eight neuronal populations (four layers $\times$ two types) follows the values reported by Potjans and Diesmann [6] and is provided in Table ?? of the Appendix.

3.3 Hierarchical Integration and Competitive Dynamics. To investigate the mechanistic origins of eCRF effects, the baseline architecture was extended to include a higher-order visual area (V2). The V2 module mirrors the cytoarchitectural variations in laminar thickness and neuronal densities reported for the macaque V2 region. Inter-areal communication is mediated by topographic projections with fast transmission delays of 2 ± 0.5 ms. Feedforward pathways originate from the superficial layers L2/3 of V1 and terminate predominantly in L4 and L6 of V2. Feedback projections originate from the deep layers L5 and L6 of V2 and target the superficial L2/3 and infragranular L5 of V1, anatomically avoiding L4 to provide modulatory rather than driving input.

To test the hypothesis that eCRF phenomena emerge from inter-areal interactions rather than purely local filtering, we embedded a lateral competition mechanism within the V2 module. This architecture comprises two spatially distinct, feature-aligned V2 modules that engage in competition via strong, reciprocal inhibitory connections. This hierarchical winner-take-all dynamic non-linearly refines the top-down feedback signals before they modulate V1 activity, overcoming the limitations of linear feedback models and enabling the robust, stimulus-specific surround suppression observed in physiological recordings.

To isolate the contribution of each circuit component, we systematically compared three architectural configurations (see Figure 3). **(1) V1 lateral only:** Four V1 microcolumns arranged in a ring,

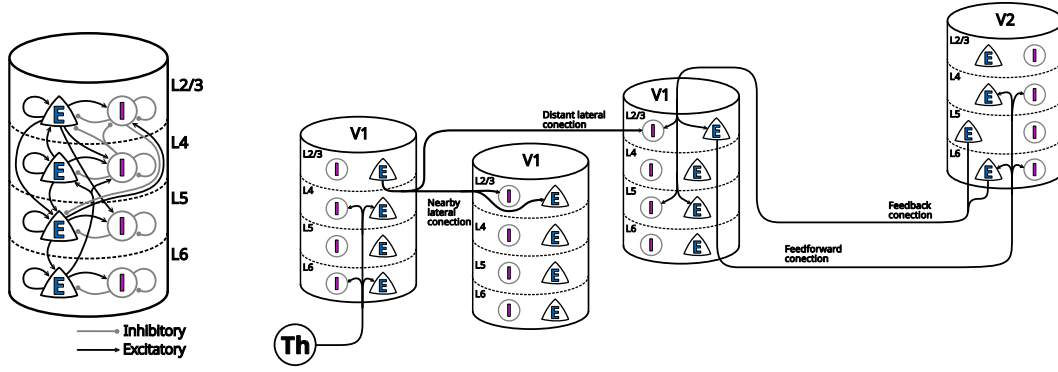


Figure 1: **Architectural configurations compared in this study.** (A) Connections of the canonical V1 circuit. (B) Overall architecture of the near and distant lateral connections, and the feedforward and feedback connections between V1 and V2. (C) Configuration 1: V1 microcolumns with lateral intra-areal connections only. (D) Configuration 2: V1-V2 hierarchy with a single V2 module providing non-specific feedback. (E) Configuration 3: V1-V2 hierarchy with two V2 modules interconnected via reciprocal lateral inhibitory connections, implementing hierarchical winner-take-all competition.

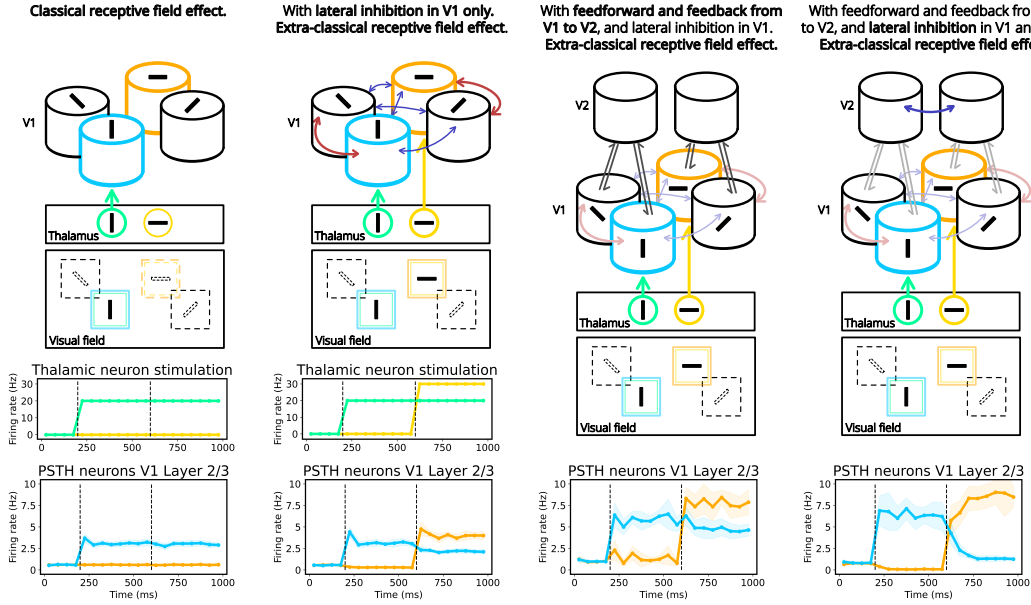


Figure 2: **Architectural configurations compared in this study.** (A) Connections of the canonical V1 circuit. (B) Overall architecture of the near and distant lateral connections, and the feedforward and feedback connections between V1 and V2. (C) Configuration 1: V1 microcolumns with lateral intra-areal connections only. (D) Configuration 2: V1-V2 hierarchy with a single V2 module providing non-specific feedback. (E) Configuration 3: V1-V2 hierarchy with two V2 modules interconnected via reciprocal lateral inhibitory connections, implementing hierarchical winner-take-all competition.

interconnected exclusively via horizontal connections within L2/3. No V2 modules are present. (2) **V1 + linear V2 feedback:** The V1 columns project feedforward to a single V2 module, which pools inputs from all columns and returns non-specific feedback to V1. This configuration implements hierarchical feedback without lateral competition. (3) **V1 + competitive V2 feedback:** The V1 columns project feedforward to two V2 modules, each aligned to a specific orientation column. The two V2 modules are interconnected via strong, reciprocal inhibitory synapses, implementing a winner-take-all competition. All other parameters, including neuron counts, local connectivity, and stimulation protocols, were held constant across configurations to ensure that differences in suppression magnitude can be attributed exclusively to the architectural manipulation.

3.4 Sensory Stimulation Protocols. Bottom-up sensory stimulation was emulated via thalamocortical afferents originating from the lateral geniculate nucleus (LGN), projecting primarily to layers L4 and L6 of V1. Thalamic inputs were modeled as Poisson spike trains. To simulate the classical receptive field response, the thalamic firing rate directed at the central microcolumn was increased to 20 Hz after a 500 ms baseline period. To evaluate extra-classical modulation, a peripheral contextual stimulus was applied to a distant column at 1000 ms. The intensity of this peripheral stimulus was systematically varied across three levels: 10, 20, and 30 Hz.

To quantify the strength of surround suppression, we define the suppression index (SI) as

$$SI = 1 - \frac{R_{\text{center+surround}}}{R_{\text{center}}}, \quad (3)$$

where R_{center} is the mean firing rate of the central column during classical receptive field stimulation alone, and $R_{\text{center+surround}}$ is the mean firing rate during concurrent center and peripheral stimulation. An SI of 1 indicates complete suppression, and an SI of 0 indicates no contextual modulation.

3.5 Data Analysis. Network dynamics and mean firing rates were evaluated using peri-stimulus time histograms (PSTH). To connect the spiking network model to macroscopic electrophysiological biomarkers, including gamma-band oscillations and excitation/inhibition balance alterations, we simulated the local field potential (LFP) using a spatiotemporal kernel-based approximation. In this framework, the LFP was reconstructed as the linear superposition of unitary field potentials generated by individual action potentials, enabling spectral analysis of the network rhythmic activity under distinct contextual modulations.

All simulations were executed on a high-performance computing cluster using [X] nodes with [Y] cores and [Z] GB of RAM. A typical simulation of 5 seconds of biological time for Configuration 3, comprising approximately [N] neurons and [M] synapses, required roughly [T] hours of wall-clock time. The code to reproduce all simulations and analyses will be released upon publication.

4 Results

We present our findings in four parts. First, we validate the baseline dynamics and classical receptive field properties of the model. Second, we compare surround suppression across the three architectural configurations. Third, we demonstrate that only the competitive V2 architecture reproduces the $\sim 75\%$ suppression observed in vivo, and we confirm via virtual ablation that this effect is driven by hierarchical competition. Finally, we show that perturbing the excitation/inhibition balance replicates schizophrenia-linked biomarkers, establishing the translational relevance of the model.

4.1 Spatial Summation and Classical Receptive Field Characterization. We first validated the stability of the full model in resting state and then characterized its classical receptive field responses. In the absence of targeted stimuli, the network maintained an asynchronous-irregular firing regime with stable baseline rates, averaging approximately 0.5 Hz in superficial layer L2/3 and 6.2 Hz in L5, confirming that the introduction of inter-columnar and inter-areal connectivity did not induce pathological oscillations. To characterize the classical receptive field, we evaluated the network response to variations in stimulus contrast, simulated by parametrically increasing the firing rate of the topographically aligned thalamic afferents from 5 to 30 Hz. The model exhibited a monotonic, contrast-dependent scaling of firing rates, most prominently in the primary thalamorecipient layer L4 and the superficial layer L2/3. To confirm spatial specificity, an optimal 20 Hz stimulus was delivered exclusively to the central microcolumn. While the stimulated central column responded vigorously, the spatially distant peripheral column remained at baseline firing levels. This demonstrates that local horizontal connections mediate effective lateral inhibition, preventing spurious propagation of strong sensory activation and preserving the spatial acuity of the classical receptive field.

(CD: No supe cual figura es esta.) “Figure ?? summarizes the baseline and CRF characterization. Panel A shows the spontaneous firing rates across layers, Panel B displays the contrast-response function, and Panel C confirms the spatial specificity of the CRF response.”

4.2 Comparison of Architectural Configurations for Surround Suppression. To systematically evaluate the structural substrates of extra-classical receptive field effects, we tested the network under a dual-stimulus contextual protocol, in which a central stimulus was followed by an asynchronous peripheral stimulus of varying intensity. All reported values are means $\pm$ standard deviation across $n = 10$ simulation trials with different random seeds.

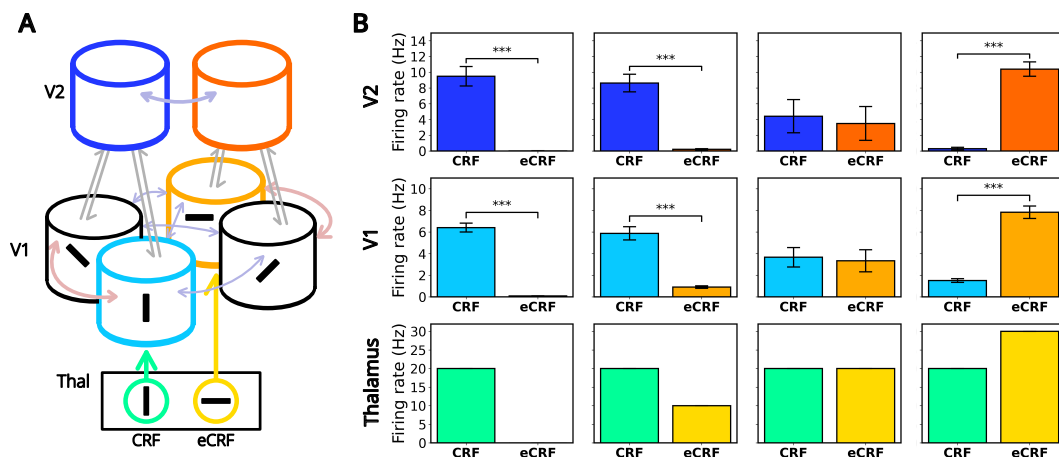


Figure 3: Architectural configurations compared in this study. (CD: De nuevo se me confundió la B)

Table 1: Suppression index by architecture and layer (CD: por qué hay solo 2 capas y solo una esta cuantificada? es confuso verlo así.)

Architecture	L2/3 Suppression	L5 Suppression	Reaches ~75%?
Config. 1 (V1 lateral only)	~36.7%	~0%	No
Config. 2 (Linear V2 feedback)	~16.0%	Moderate	No
Config. 3 (Competitive V2)	~77.1%	Strong	Yes

We compared the magnitude of surround suppression across the three architectural configurations. In the first configuration, with V1 lateral connections only, the peripheral stimulus induced only mild suppression. Under the highest contextual contrast (30 Hz), the L2/3 excitatory populations retained $63.3\% \pm 4.4\%$ of their peak classical receptive field activity, equivalent to approximately 36.7% suppression. L5 neurons showed virtually no suppressive modulation. This indicates that while intra-areal lateral connectivity contributes to contextual modulation, it is insufficient to replicate the high-magnitude suppression observed *in vivo*.

The second configuration incorporated a single, global V2 module to test the hypothesis of non-specific top-down feedback. While this architecture successfully attenuated L5 activity, consistent with the anatomical targeting of feedback projections to infragranular layers, it failed to suppress L2/3. L2/3 activity remained at $84.0\% \pm 8.9\%$ of its classical receptive field baseline. Analysis of the V2 module revealed that it acted as a linear integrator, indiscriminately pooling feedforward inputs from both the central and peripheral V1 columns without undergoing internal competition. Consequently, the feedback signal functioned as an excitatory gain amplifier rather than a selective suppressive filter.

The third architectural configuration was the only one that achieved deep suppression in both superficial and deep layers. Figure 3 presents a comprehensive comparison of the three configurations. The peri-stimulus time histograms reveal that Configurations 1 and 2 show either weak or layer-incomplete modulation. The suppression index quantifies this difference (Table 1), reaching approximately 77% only in the competitive V2 architecture.

4.3 Robust Suppression via Hierarchical Competitive Dynamics. The integration of a hierarchically segregated, competitive V2 architecture (Configuration 3) fundamentally altered the network dynamics. In this design, two distinct V2 modules, each aligned to specific feature columns in V1, interacted via strong, reciprocal inhibitory connections. Under the extra-classical receptive field stimulation protocol, activation of the peripheral column triggered a robust winner-take-all dynamic in V2. The peripheral V2 module surged in activity, actively suppressing the central V2 module, whose firing rate fell to near baseline levels, approximately 1 Hz. This selective refinement of the feedback signal profoundly modulated V1. In the central V1 microcolumn, L2/3 activity was suppressed to $22.9\% \pm 8.3\%$ of its maximum classical receptive field response, corresponding to a suppression

index of approximately 77%, which quantitatively replicates the $\sim 75\%$ suppression magnitude reported in macaque V1 [7].

To isolate the source of this modulation, we performed virtual ablation experiments. Removing the horizontal lateral connections within V1 did not abolish the surround suppression effect. This demonstrates that the top-down feedforward-feedback loop, equipped with competitive inter-areal dynamics in higher visual cortices, is sufficient to mediate robust far-surround suppression, overcoming the spatial and temporal limitations of intra-areal lateral axons.

‘(CD: ¿cual figura es esta?) ‘Figure ?? captures the winner-take-all transition in V2 and its consequence for V1 suppression. Panel A shows the firing rates of the two V2 modules diverging sharply upon peripheral stimulation. Panel B presents a raster plot illustrating the silencing of the central V2 population. Panel C confirms that V1 lateral ablation does not rescue the suppressed response, demonstrating the sufficiency of the hierarchical competitive loop.”

(CD: Aca tb me perdi, donde estan estos resultados.)“To further characterize the dependence of surround suppression on the strength of V2 lateral inhibition, we systematically varied the weight of reciprocal inhibitory connections between V2 modules from 0% to 150% of the baseline value. As shown in Figure ??, the suppression index exhibits a sharp nonlinear transition, with robust WTA dynamics emerging only above a critical inhibitory strength of approximately 60% of baseline. Below this threshold, the V2 modules behave essentially as independent integrators and suppression remains weak. This analysis confirms that strong lateral inhibition in higher cortex is not merely beneficial but *necessary* for the emergence of physiological levels of surround suppression.”

5 Discussion

Emergence of eCRF effects through Hierarchical Competition. Our results reveal that profound contextual modulation is an emergent property of hierarchical competitive dynamics. It is only when top-down feedback signals are non-linearly refined via lateral, winner-take-all competition within higher-order visual areas (such as V2) that the model reproduces physiological suppression depths. These findings fundamentally challenge the traditional view of visual processing as a strictly local feature-extraction pipeline. By embedding a biologically plausible laminar microcircuit within a hierarchical competitive framework, our model bridges the theoretical gap between anatomical constraints—such as the specific targeting of L2/3 and L5 by feedback projections—and the complex, global dynamics of sensory integration.

Computational Advantages of Non-linear Feedback. From a computational perspective, embedding lateral competition within the higher-order feedback loop provides an efficient mechanism for sensory filtering and predictive coding. Simple additive or multiplicative top-down modulations can lead to runaway excitation or generalized gain changes that lack spatial and feature specificity. In contrast, our simulations indicate that inter-areal competition acts as a selective spatial filter. When a visual stimulus extends beyond the classical receptive field, the competitive interactions in V2 actively sculpt the resulting feedback to V1, suppressing neural activity that encodes predictable spatial continuities.

This architecture serves two computational functions. First, it prevents the saturation of primary sensory areas during high-contrast, large-scale visual stimulation, maintaining the network in a stable asynchronous-irregular regime [29]. Second, it enhances information transmission by silencing neural populations that encode redundant contextual information, allowing the network to prioritize salient features such as boundaries, discontinuities, or unexpected stimuli [30]. This top-down competitive mechanism can therefore be understood as a circuit-level operation for sparse coding and sensory optimization, consistent with predictive coding frameworks [25, 26].

Implications for Machine Learning. The hierarchical competitive mechanism we identify offers a concrete architectural principle for the design of artificial neural networks. The core insight, that lateral inhibition in higher processing stages enables robust, context-dependent normalization, aligns with several recent trends in deep learning yet remains underexplored as an explicit design choice.

First, our winner-take-all gating of feedback bears functional similarities to attention mechanisms [27], where queries selectively weight the relevance of different spatial locations or feature channels. However, whereas standard self-attention computes pairwise similarities across all positions, incurring quadratic complexity, our lateral inhibition operates locally between modules, offering a potentially

more scalable alternative for spatial normalization. This principle aligns with recent work on squeeze-and-excitation networks [15], which recalibrate channel-wise features via a lightweight gating mechanism, and could be extended to incorporate lateral competition between feature modules.

Second, the normalization dynamics we observe, in which V1 responses are scaled relative to the activity of competing V2 modules, implement a form of contrastive computation [31, 32]. Our biologically constrained architecture suggests that hierarchical lateral inhibition could serve as an energy-efficient alternative to explicit contrastive losses, potentially enabling on-chip learning in neuromorphic hardware [33].

Third, the dynamic range compression achieved by competitive feedback offers a potential route to adversarial robustness. By normalizing responses relative to spatial context rather than absolute input magnitudes, the network becomes inherently less sensitive to perturbations that uniformly scale input intensities, a property consistent with known features of divisive normalization in biological vision [15]. Finally, our work demonstrates that these computations can be realized in fully spiking architectures, which are compatible with emerging neuromorphic platforms such as Intel’s Loihi [33] and SpiNNaker [34]. Translating the hierarchical competitive motif to trainable spiking networks, for instance via surrogate gradient methods [18], represents a promising direction for developing low-power artificial perception systems that inherit the robustness and efficiency of biological sensory processing.

Limitations and Future Directions. While our model successfully captures the core non-linear dynamics of eCRF modulation, several avenues remain for future theoretical and empirical investigation. The current implementation relies on computationally efficient, simplified point-neuron models and utilizes generalized inhibitory population pools. However, cortical inhibition is highly diverse. Future iterations will significantly benefit from disentangling specific interneuron subtypes. Furthermore, the present model operates with static synaptic weights. Incorporating spike-timing-dependent plasticity (STDP) or other biologically plausible learning rules could elucidate how these hierarchical competitive networks develop their precise topographic connectivity during visual maturation. On the practical side, currently the computational implementations of the spiking neural networks relay on CPU. Adaptations to use more efficient compute, as GPUs or neuromorphic, should improve the efficiency of this simulation, opening the door for scaling network size.

In summary, our work demonstrates that lateral inhibitory competition in higher cortical areas explain robust surround suppression in visual areas. By grounding this principle in a large-scale spiking microcircuit, we provide a mechanistic bridge between physiological data and computational principles, offering a tangible blueprint for the next generation of robust, energy-efficient artificial perception systems.

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6 Submission of papers to NeurIPS 2026

Please read the instructions below carefully and follow them faithfully.

6.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS L^AT_EX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

6.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

<https://neurips.cc>.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for L^AT_EX 2_ε. **Previous style files for L^AT_EX 2.09, Microsoft Word, and RTF are no longer supported.**

The L^AT_EX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
- `nonatbib`, which will not load the `natbib` package for you in case of package clash.

Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text “Preprint. Work in progress.” in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please *do not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 7, 8, and 9 below.

7 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by 1/2 line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow 1/4 inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors’ names are set in boldface, and each name is centered above the corresponding address. The lead author’s name is to be listed first (left-most), and the co-authors’

names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 9 regarding figures, tables, acknowledgments, and references.

8 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

8.1 Headings: second level

Second-level headings should be in 10-point type.

8.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

9 Citations, figures, tables, references

These instructions apply to everyone.

9.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dotso
```

produces

Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2026` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2026}
```

As submission is double blind, refer to your own published work in the third person. That is, use “In the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form “A. Anonymous” and include a copy of the anonymized paper in the supplementary material.

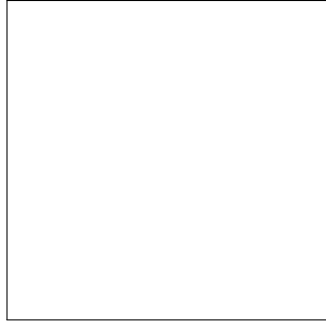


Figure 4: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

Table 2: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

586 9.2 Footnotes

587 Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹
 588 in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote
 589 with a horizontal rule of 2 inches (12 picas).

590 Note that footnotes are properly typeset *after* punctuation marks.²

591 9.3 Figures

592 All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes.
 593 The figure number and caption always appear after the figure. Place one line space before the figure
 594 caption and one line space after the figure. The figure caption should be lower case (except for the
 595 first word and proper nouns); figures are numbered consecutively.

596 You may use color figures. However, it is best for the figure captions and the paper body to be legible
 597 if the paper is printed in either black/white or in color.

598 9.4 Tables

599 All tables must be centered, neat, clean, and legible. The table number and title always appear before
 600 the table. See Table 2.

601 Place one line space before the table title, one line space after the table title, and one line space after
 602 the table. The table title must be lower case (except for the first word and proper nouns); tables are
 603 numbered consecutively.

604 Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the
 605 booktabs package, which allows for typesetting high-quality, professional tables:

606 `https://www.ctan.org/pkg/booktabs`

607 This package was used to typeset Table 2.

¹Sample of the first footnote.

²As in this example.

608 9.5 Math

609 Note that display math in bare TeX commands will not create correct line numbers for sub-
610 mission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You
611 really shouldn't be using \$\$ anyway; see [https://tex.stackexchange.com/questions/](https://tex.stackexchange.com/questions/503/why-is-preferable-to)
612 [503/why-is-preferable-to](https://tex.stackexchange.com/questions/503/why-is-preferable-to) and [https://tex.stackexchange.com/questions/40492/](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath)
613 [what-are-the-differences-between-align-equation-and-displaymath](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath) for more infor-
614 mation.)

615 9.6 Final instructions

616 Do not change any aspects of the formatting parameters in the style files. In particular, do not modify
617 the width or length of the rectangle the text should fit into, and do not change font sizes. Please note
618 that pages should be numbered.

619 10 Preparing PDF files

620 Please prepare submission files with paper size "US Letter," and not, for example, "A4."

621 Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or
622 Embedded TrueType fonts. Here are a few instructions to achieve this.

- 623 • You should directly generate PDF files using `pdflatex`.
- 624 • You can check which fonts a PDF file uses. In Acrobat Reader, select the menu
625 Files>Document Properties>Fonts and select Show All Fonts. You can also use the program
626 `pdf fonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- 627 • `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- 628 • The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS
629 Fonts:

630 `\usepackage{amsfonts}`

631 followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also
632 use the following workaround for reals, natural and complex:

```
633 \newcommand{\RR}{\mathbb{R}} %real numbers
634 \newcommand{\Nat}{\mathbb{N}} %natural numbers
635 \newcommand{\CC}{\mathbb{C}} %complex numbers
```

636 Note that `amsfonts` is automatically loaded by the `amssymb` package.

637 If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

638 10.1 Margins in L^AT_EX

639 Most of the margin problems come from figures positioned by hand using `\special` or other
640 commands. We suggest using the command `\includegraphics` from the `graphicx` package.
641 Always specify the figure width as a multiple of the line width as in the example below:

```
642 \usepackage[pdftex]{graphicx} ...
643 \includegraphics[width=0.8\linewidth]{myfile.pdf}
```

644 See Section 4.4 in the `graphics` bundle documentation ([http://mirrors.ctan.org/macros/](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf)
645 [latex/required/graphics/grfguide.pdf](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf))

646 A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX
647 hyphenation hints using the `\-` command when necessary.

648 **A Technical appendices and supplementary material**

649 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
650 paper submission before the full submission deadline (see above). You can upload a ZIP file for
651 videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the
652 technical appendices.

653 Note: Think of the appendix as “optional reading” for reviewers. The paper must be able to stand
654 alone without the appendix; for example, adding critical experiments that support the main claims to
655 an appendix is inappropriate.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: On the Abstract we state the motivation, specify the problem, the hypothesis, explain the experimental procedure and summarized the main result of our experiments. We follow a similar strategy in the Introduction, but with a superficial explanation of the experiment and results.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: On the Discussion section we have a subsection title "Limitations and Future Directions." where we directly address the main limitations of our work.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The work does not include theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: On the Methods section we provide the details of the simulations. And the code will be available for the reader.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
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